

Quantitative Geography Showcase

GEOG 531: Quantitative Methods in Geographic Research | Fall 2026

Final Showcase: Monday, December 8, 4:00–7:00 PM

Jones Physical Science Building Room 102a

What is the Showcase?

The **Quantitative Geography Showcase** is your opportunity to tell a clear, evidence-based geographic story with data. You will pose a focused question, carry out a reproducible analysis, communicate the results to a real audience, and explain both what your evidence supports and what it does *not* support.

This is not a contest for the most elaborate model or the largest dataset. It is about communicating something meaningful about a geographic pattern, relationship, or process using class tools and concepts. Let your imagination and creativity run wild!!

The Showcase is worth **150 points (15% of your grade)**. You may choose your own geographic research question and data, or work on one given by your instructor. Work must be completed individually, unless you would like to work with another class member. If so, the scope of your project will be adjusted accordingly in consultation with your instructor

Because this is a small class, the final session will be a **mini quantitative geography conference** rather than a large, boring presentation run. Every student will present to—and receive thoughtful questions from—the full class. The aim is to leave the session with sharper ideas, useful feedback, and a better understanding of the varied geographic questions our class can address.

Learning goals

By completing the Showcase, you will practice how quantitative geographic research moves from a question to an interpretable conclusion. In particular, you will:

- frame a geographic question that can be investigated with available data;
- use descriptive statistics, maps, and visualizations to understand the data;
- conduct exploratory spatial or multivariate analysis to identify patterns and guide decisions;
- use an appropriate inferential analysis to evaluate a relationship, difference, or hypothesis;

- communicate findings honestly, including uncertainty, assumptions, and limitations; and
- produce work that another person can inspect and reproduce.

Choose a question worth answering

Start with a question—not with a method. Your question should name a phenomenon, a place or spatial unit, a population or time period, and a relationship or pattern you want to investigate.

Good projects are usually narrow enough to answer within a semester. For example:

- How does access to public transit vary across neighborhoods in a metropolitan area, and is it associated with neighborhood socioeconomic characteristics?
- Are heat-related emergency visits clustered in particular counties, and what county characteristics are associated with higher rates?
- Do park amenities and population density predict park visitation across census tracts?
- How do housing costs differ across neighborhoods, and is proximity to employment centers related to those differences?
- Are reported air-quality values associated with neighborhood characteristics after accounting for spatial patterning?

You are welcome to work on a topic relevant to your major, community, workplace, or interests. You do **not** need an original discovery. A well-executed analysis of a familiar question is preferable to an overly ambitious question that cannot be answered responsibly.

Is my project doable?

Before committing to a question, make sure you can answer “yes” to all of these:

- Can I obtain or create an analyzable dataset with a clear unit of observation?
- Do I understand what each key variable measures and its limitations?
- Can I define a meaningful study area and time period?
- Can I complete the core analysis with the methods taught in this course?
- Can I explain why my results should matter to a specific audience?

Required analytical components

Your project should follow a logical research workflow. The components below are required, but the exact techniques should fit your question and data. Do not add an analysis simply to “check a box.” Explain why each decision helps answer your question.

Component	What your project must show	Possible approaches
1. Geographic research question & context	A focused question, a brief explanation of why it matters, the study area, and a clear unit of analysis.	Short literature/context review; study-area map; conceptual diagram; data dictionary.
2. Data & study design	Data source(s), date or period, inclusion/exclusion decisions, variable definitions, and important data-quality concerns.	Public datasets, administrative data, field observations, surveys, APIs, etc..
3. Descriptive spatial statistics	Numerical and visual summaries that establish what the data look like before formal modeling.	Counts, percentages, rates, central tendency, spread, distribution plots, choropleths, dot maps, standard distance, or other appropriate summaries.
4. Data visualization	At least two purposeful, polished visualizations; at least one must be a map when geography is central to the question.	Map plus distribution plot, scatterplot, small multiples, coefficient plot, interactive figure, or panel maps.
5. Exploratory analysis	An examination of patterns, relationships, heterogeneity, or potential problems that informs the next analytical step.	Cross-tabs, scatterplots, correlation, point/area pattern analysis, Moran's I/LISA.
6. Inferential analysis	One appropriate inferential method connected directly to the research question, with an interpretation beyond "statistically significant."	Confidence interval, permutation or simulation-based test, comparison of groups, correlation, linear/GLM regression, multivariable regression, spatial regression, or another instructor-approved method.
7. Interpretation, uncertainty & limits	A conclusion that distinguishes description, association, prediction, and causation; discusses uncertainty and at least two limitations.	Model diagnostics, sensitivity analysis, discussion of sampling, measurement, aggregation, ecological fallacy, missingness, or spatial dependence.

Minimum analytical expectation

At a minimum, your analysis must include **descriptive spatial statistics**, **purposeful visualization**, **exploratory analysis**, and **one inferential analysis**. Your results should be presented as a connected story: descriptive and exploratory work should motivate

the inferential step, and the inferential result should be interpreted in light of the geographic context.

Use the simplest defensible method. A straightforward test or regression that is clearly justified and well interpreted will score better than a complex analysis that is poorly explained or unsupported by the data.

Deliverables

Submit the following items as instructed on the course site. Your analysis must be your own work and must comply with the course AI and academic-integrity policies.

1. Reproducible analysis

You will submit a rendered **Quarto document** (HTML or PDF, as specified) and the associated .qmd file in your class posit folder as part of a final geography showcase assignment. Your analysis should run from raw or clearly documented input data to final results.

Your document should include:

- a concise introduction and research question;
- data sources, spatial unit, study area, and data-preparation decisions;
- commented code organized into logical sections;
- tables and figures that are numbered, titled, and interpreted in the text;
- a description of the analytical methods and why they are appropriate;
- results and diagnostics/checks relevant to your method; and
- a short conclusion and references.

Reproducibility standard: another student should be able to follow the code, identify the input data, and understand how each major result was produced. Your code does not need to be perfect or elegant, but it must be readable and should not contain unexplained manual steps. Cite all data, code, maps, and outside sources.

2. Audience-facing communication product

Create one polished product that communicates your main finding to a real, non-specialist or mixed audience. Choose a format that fits your topic:

- a one-page research brief or policy memo;
- a poster or infographic;
- a short story map or web-based visual narrative;
- a news-style data story; or
- another format approved by the instructor.

Your product should make the central takeaway clear in the first few seconds. It should use a strong title, readable visual hierarchy, carefully selected figures/maps, plain-

language interpretation, and source attribution. It should not be a miniature copy of your full Quarto report.

3. Showcase presentation (20 points)

Give a concise **5-minute research story**, followed by approximately **3 minutes of questions and discussion**. Use slides, your poster, a story map, or another visual format that helps the audience follow your story. Your audience will include classmates with different topical interests but shared training in quantitative geography, so make your decisions understandable rather than assuming they are obvious. The instructor may also invite other students/professors to attend based on interest.

Your presentation should answer five questions:

1. What geographic question did you investigate, and why should we care?
2. What data and study area did you use?
3. What is the key spatial pattern or descriptive finding?
4. What did your inferential analysis show?
5. What should an audience conclude—and what remains uncertain?

Practice for time. A concise, well-paced story with one or two strong visuals is more effective than trying to show every figure or line of code.

4. Critical reflection

Submit a **500–700 word reflection**. Address all of the following:

- What is the most important thing your analysis can support?
- What is one conclusion that would be tempting but unjustified?
- Which analytical decision was most consequential, and why?
- What are two meaningful limitations or sources of uncertainty?
- If you had more time, data, or expertise, what would you do next?

Project development is built into Homework 2 and Homework 3

You will not be asked to invent the project all at once near the end of the semester. **Homework 2 and Homework 3 will each include a small set of questions about your emerging final project**, for a few points within those homework assignments. These prompts will ask you to think concretely about the following:

Course point	What you will develop	Why it matters
Homework 2: exploratory analysis	A provisional research question; study area and unit of analysis; possible data source(s); one exploratory map, table, or figure; and a first observation or question raised by the data.	It tests whether the idea is feasible and helps you discover the geographic pattern before selecting a model.

Course point	What you will develop	Why it matters
Homework 3: inferential analysis	A refined question; key outcome and explanatory variables; a proposed inferential method; a brief justification; and one assumption, limitation, or diagnostic you will need to consider.	It aligns the method with the question and creates time to revise the plan before the Showcase.

You may change your topic after receiving feedback. The purpose of these questions is to make your final project more manageable and more thoughtful, not to lock you into an early idea. In-class work time and peer consultation will also be available as the project develops.

The December 8 mini-conference

The final session (4:00–7:00 PM) will be structured as a small, supportive research conference. Ten students means that every project can receive real attention rather than a quick pass-by.

Part of the session	How it will work
Research stories	Each student gives a 5-minute presentation followed by about 3 minutes of questions. Everyone will hear every presentation.
Showcase conversation	Bring or display your audience-facing product. Classmates will have time to look closely, ask follow-up questions, and talk through choices in data, mapping, and methods.
Constructive peer feedback	Each student will offer at least one substantive question and one useful piece of feedback to a classmate. The best questions are specific: they push a claim, interpretation, scale choice, or next step forward.
Closing reflection	We will end by sharing one methodological lesson or idea from another project that you want to carry into future work.

Make the project yours. A compelling title, a surprising or carefully explained finding, and an audience-aware communication product are encouraged. Creativity is welcome when it clarifies the analysis—not when it obscures it. Your maps and graphics should be attractive, accessible, and honest about the data.

Evaluation rubric (150 points)

Criterion	Excellent work	Points
Research question, geographic framing & data	Focused, feasible, and meaningful question; clear study area/unit of analysis; high-quality, well-documented data; thoughtful attention to scale and data limitations.	20

Criterion	Excellent work	Points
Descriptive statistics & visualization	Accurate, insightful summaries; polished maps/figures that reveal patterns and are designed for their intended audience; all visuals are labeled and interpreted.	25
Exploratory analysis & methodological reasoning	Exploratory work meaningfully guides the analysis; methods are appropriate and clearly justified; key choices and alternatives are considered.	20
Inferential analysis & interpretation	Correct, transparent analysis; assumptions/diagnostics addressed as appropriate; effect sizes, uncertainty, and geographic meaning interpreted accurately.	25
Reproducibility & technical communication	Quarto analysis is organized, readable, and runs with documented inputs; sources, code, and data are properly cited.	20
Audience-facing product & presentation	Clear, engaging, and professionally designed; communicates a coherent research story within the time limit; contributes substantive questions and feedback to peers.	20
Critical reflection	Honest, specific reflection on limitations, uncertainty, and next steps.	20
Total		150

Expectations for undergraduate and graduate students

All students complete the same core Showcase and are evaluated on the same 150-point rubric. The difference is not that graduate students must use a more complicated method; it is that they must offer a more explicit and critical account of their methodological decisions.

Dimension	Undergraduate students	Graduate students
Research context	Explain why the question matters and cite data/other sources used.	Situate the question in relevant scholarly literature and explain the project's contribution or connection to that work.
Method choice	Select and explain an appropriate course method.	Justify the selected method relative to at least one plausible alternative, including why the alternative was not preferred or what different inference it would support.
Checks and uncertainty	Identify appropriate limitations and complete basic diagnostic or	Include at least one substantive robustness, sensitivity, or diagnostic exercise beyond the basic analysis,

Dimension	Undergraduate students	Graduate students
	assumption checks when relevant.	and explain how it changes—or does not change—the interpretation.
Inference	Interpret results accurately; distinguish association from causation when needed.	Explain how data structure, scale, spatial dependence, model specification, or other methodological choices shape the inference.
Showcase discussion	Respond to questions about the research story.	Be prepared to briefly defend a methodological choice and discuss a reasonable next analytical step.

Graduate-level rigor is about transparency, comparison, and reflection—not about using advanced techniques for their own sake. Seniors who want to attempt these extensions are encouraged to do so with instructor guidance.

Practical guidance

Keep it manageable. A single well-defined study area and one strong question are enough. Start with data you can access now.

Let the data surprise you. Exploratory analysis is not a formality. If your first map or plot changes the question, that is often good research.

Do not overclaim. An association is not automatically a causal effect. A clustered map does not, by itself, identify a cause. Be precise about your evidence.

Use maps carefully. Choose projection, classification, color, scale, and labels intentionally. Make sure a map does analytical work rather than serving as decoration.

Ask for feedback early. Bring incomplete work to workshops. The best time to discover a problem with a dataset or method is before the final week.

Plan for a real audience. Imagine who could use your work (residents, a local agency, a nonprofit, planners, researchers, or classmates) and design your communication product for them.

Final pre-submission checklist

- ☐ My research question is specific and geographic.
- ☐ I identify the study area, spatial unit, period, and data sources.
- ☐ I include descriptive spatial statistics and purposeful visualizations.
- ☐ I conduct exploratory analysis that informs my next step.
- ☐ I use and justify an inferential analysis appropriate to my question.
- ☐ I interpret findings with uncertainty and avoid unsupported causal claims.
- ☐ My Quarto file and rendered output are readable and reproducible.

- ☐ My communication product tells a clear story for a real audience.
- ☐ My reflection states what the analysis can and cannot support.
- ☐ I cite data, software, maps, code, and outside sources as needed.